

Rounding, Estimating, Ordering

Name: _____

IGCSE 0607 Extended Mathematics Practice Questions
prepared by Steve Cheung

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- 1 (a) Write 1234.5 correct to 2 significant figures.

..... [1]

- (b) By writing each number correct to 1 significant figure, estimate the value of

$$\frac{3.14 \times 19.8^2}{4.1}$$

..... [2]

2 (a) Write 45.826 correct to

(i) 1 decimal place,

..... [1]

(ii) 2 significant figures.

..... [1]

(b) By writing each number correct to 1 significant figure, estimate the value of

$$\frac{39.8 \times 2.15}{0.48}$$

..... [2]

3 Round each given value according to the specified limits of accuracy.

(a) Write 0.003 049 correct to 2 significant figures.

..... [1]

(b) Write 79 512 correct to 1 significant figure.

..... [1]

- 4** (a) Write 0.07635 correct to 3 decimal places.

..... [1]

- (b) Estimate the value of the following expression by writing each number correct to 1 significant figure.

$$\frac{\sqrt{98.5} \times 4.1}{1.93}$$

..... [2]

- 5** Write the number 483.7062 correct to the specified degrees of accuracy.

- (a) Correct to 2 decimal places.

..... [1]

- (b) Correct to 3 significant figures.

..... [1]

- 6 (a) Arrange the following real terms in ascending order of magnitude.

$$\frac{17}{25}, \quad 67.5\%, \quad 0.682, \quad \left(\frac{1}{8}\right)^{-1} \times 10^{-1}$$

..... [2]

- (b) Complete the expression using the correct symbol from the set $\{>, <, \geq, \leq, =\}$.

$$3.5\% \quad \text{-----} \quad \frac{7}{200}$$

..... [1]

- 7 The following four quantities are written in different numerical formats:

$$A = \frac{17}{20}, \quad B = 84.5\%, \quad C = 0.844, \quad D = \sqrt{\frac{18}{25}}$$

Use your graphics calculator to convert these values.

- (a) Write down the value of D correct to 4 decimal places.

..... [1]

- (b) Arrange the four quantities A , B , C , and D in order of magnitude, starting with the smallest.

..... [2]

- (c) Complete the statement using one of the relational symbols $=$, $\neq$, $>$, or $<$ to make it correct.

$$A \text{ ----- } \frac{85}{100}$$

..... [1]

- 8 (a) Arrange the following list of rational numbers in ascending order of magnitude.

$$0.42, \quad \frac{2}{5}, \quad 45\%, \quad \frac{9}{20}$$

..... [2]

- (b) Insert the correct relational symbol ($>$ or $<$ or $=$) into the statement below to make it true.

$$-\frac{3}{4} \quad \text{-----} \quad -0.8$$

..... [1]

- 9 Five numbers are positioned along an interval of a real number line:

$$K = 0.63, \quad L = \frac{14}{22}, \quad M = 63.5\%, \quad N = \sqrt{0.4}, \quad P = \frac{\pi}{5}$$

- (a) Calculate the value of P correct to 4 decimal places.

..... [1]

- (b) Order the five numbers from smallest to largest magnitude.

..... [2]

- (c) Complete the statement below using the most appropriate relational symbol from the set $\{<, =, >\}$.

L _____ M

..... [1]